

Distinguishing a Local Boost from a Global Tilt

The Statistical Formalism of the BASS/HTT Low- ℓ Pipeline,
and its Response to the “Leaky Universe” Test

BASS/HTT programme — Soongsil OMEG Institute

2026-07-01

Abstract

This note is a self-contained, first-principles description of how the current code separates a *local kinematic boost* (the observer’s peculiar motion, an $\ell = 1$ Doppler/aberration effect) from a *global tilt* (a cosmological bulk flow, the Ω_{tilt} sector of the anisotropy comparator $g = (\Sigma^2, W^2, \Omega_{\text{tilt}}, \Omega_k)$). It states the observables, the leading-order response map that the pipeline actually uses, and — crucially — the second-order (β^2) coupling by which a boost leaks power into the quadrupole and could masquerade as a shear Σ^2 . It then explains the five concrete safeguards already in the code, shows why the comparator covariance is *not* diagonal beyond leading order, and gives the precise form of the two formalism upgrades (a kinematic projection operator and a coupled Fisher matrix) that would deproject the leakage. The document is written so that the “Leaky Universe” toy test (§7) can be answered with justification. All content is diagnostic-only: no Bianchi family is identified, no geometry is detected, and no shear amplitude is reported as measured.

1 The diagnostic vector and its two velocity-like sectors

The anisotropy comparator is the graded vector

$$g = (\Sigma^2, W^2, \Omega_{\text{tilt}}, \Omega_k) \in \mathbb{R}^4, \quad (1)$$

carrying the shear, vorticity, tilt, and anisotropic-curvature sectors (`http://obsstat/egs3_graded_comparator.py`). Two of these are directly velocity-like and are the subject of this note:

- Ω_{tilt} — the **global tilt**: the peculiar velocity of the matter frame relative to the CMB frame, i.e. a cosmological *bulk flow*. It is a genuine large-scale degree of freedom.
- the **local boost** $\beta = \mathbf{v}_{\text{obs}}/c$ — the *observer’s* own peculiar motion. This is not a sector of g ; it is a frame effect. Its leading imprint is the CMB kinematic dipole ($\ell = 1$).

The central difficulty is that both are “a velocity in a direction $\hat{\mathbf{v}}$ ”, so their sky imprints overlap. The pipeline’s job is to keep them distinct.

2 Observables and the leading-order response map

The low- ℓ CMB channel is the set of temperature multipole coefficients $a_{\ell m}$ for $\ell \geq 2$; the peculiar-velocity channel is the CF4 radial velocity field. The code encodes the *leading-order* sensitivity of each observable channel to each sector as a $\{0, 1\}$ support pattern D (`egs3_graded_comparator.py: _RESPONSE_SUPPORT`), Table 1.

At this (leading, EGS) order the map is *block-diagonal in the two velocity sectors*: the quadrupole responds to Σ^2 only, and both dipole channels respond to Ω_{tilt} only. The reachable subspace is rank 2 ($\{\Sigma^2, \Omega_{\text{tilt}}\}$); W^2 and Ω_k are the two null sectors. This is exactly the separation the toy test interrogates: is the “quadrupole = shear” assumption safe once the observer moves?

Table 1: The leading-EGS-order response map D . Entry 1: the channel responds to that sector at leading order; 0: structurally blind.

channel \ sector	Σ^2	W^2	Ω_{tilt}	Ω_k
CMB quadrupole ($\ell=2$)	1	0	0	0
CMB dipole ($\ell=1$)	0	0	1	0
radial-velocity dipole	0	0	1	0

Honest scope of D . The code labels D as a documented *leading-EGS-order* support pattern, and states in-module that “the exact response amplitudes need the covariant transfer”. The block-diagonal structure of Table 1 is therefore an $O(\beta^1)$ statement, not an exact one. The next section makes the $O(\beta^2)$ correction explicit.

3 The leakage: a boost sources a quadrupole

A boost of the observer by β transforms a blackbody monopole T_0 . With $\mu = \hat{\mathbf{n}} \cdot \hat{\mathbf{v}}$ the Doppler/aberration expansion of the observed temperature is

$$\frac{\Delta T(\hat{\mathbf{n}})}{T_0} = \beta \mu + \beta^2 \left(\mu^2 - \frac{1}{3} \right) \cdot \mathcal{O}(1) + O(\beta^3). \quad (2)$$

The first term is the dipole ($\ell = 1$, order β). The second term contains a **kinematic quadrupole**, because $\mu^2 = \frac{2}{3}P_2(\mu) + \frac{1}{3}$: a boost necessarily deposits an $\ell = 2$ signal of amplitude

$$a_{2m}^{\text{kin}} \sim \beta^2 T_0, \quad C_2^{\text{kin}} \sim (\beta^2 T_0)^2, \quad (3)$$

aligned with the boost axis $\hat{\mathbf{v}}$. Numerically, for $\beta \simeq 10^{-3}$ (a 300 km s^{-1} motion) one has $\beta^2 \simeq 10^{-6}$ and $\beta^2 T_0 \simeq$ a few μK , i.e. C_2^{kin} of order a few μK^2 — small against the observed $C_2 \sim \mathcal{O}(10^2) \mu\text{K}^2$, but *nonzero, sign-definite, and axis-aligned*.

Why this breaks the diagonal. Because the same velocity that sets the dipole (hence Ω_{tilt}) also sets this quadrupole, the true response map has an extra column entry that Table 1 omits:

$$D_{\text{quad}, \Omega_{\text{tilt}}} \propto \beta^2 \neq 0. \quad (4)$$

The exact D is therefore *not* block-diagonal, and the comparator covariance / Fisher matrix inherits a nonzero off-diagonal $F_{\Sigma^2, \Omega_{\text{tilt}}}$. A naive estimator that reads all quadrupole power as Σ^2 will report a false positive $\Sigma^2 > 0$ in a universe whose only anisotropy is a bulk flow. *This is the correct and important content of the toy test.*

4 How the current pipeline keeps the two apart

The code does not yet deproject (4) inside the low- ℓ statistics (there is no kinematic-quadrupole subtraction in the OBSSTAT low- ℓ path; the $O(\beta)$ Challinor aberration kernel in `htt/bass/hierarchy/boost_kernel.py` lives on the Boltzmann source side, not here). Instead, the separation is enforced by five structural safeguards, so that the leakage cannot become a false *measurement*:

1. **The dipole is excluded** ($\ell_{\text{min}} = 2$). All morphology statistics are computed on $\ell \in [2, \ell_{\text{max}}]$ (`make_lowell_morphology_real_map.py:ELL_MIN=2`). The dominant $O(\beta)$ boost imprint never enters the statistics directly; only the much smaller $O(\beta^2)$ quadrupole can leak.

2. **No shear amplitude is ever reported as measured.** Σ^2 is held at partial (K1 is a look-elsewhere descriptor under a Λ CDM null; see `pr08_006_joint_artifact.json`). The pipeline returns rank and look-elsewhere p -values, not a shear amplitude, so a β^2 leak cannot be promoted into a “detected shear”. The blind/partial sectors *fail closed*; they are never set from the quadrupole alone.
3. **A dedicated boost-alignment statistic.** One of the six registered statistics is `axis_to_cmb_dipole_deg`: the angle between the low- ℓ preferred axis and the *CMB kinematic dipole apex*. A boost-induced quadrupole is aligned with $\hat{\mathbf{v}}$ by construction, so it shows up precisely as an anomalously small axis-to-apex angle. The pipeline therefore *measures the very signature* the leakage would produce, and calibrates it against the null — rather than silently absorbing it into Σ^2 .
4. **The tilt is anchored by an independent channel.** Ω_{tilt} is measured from the CF4 radial peculiar velocities (K5 bulk flow, $|\mathbf{B}| \simeq 341 \pm 102 \text{ km s}^{-1}$), *not* from the CMB quadrupole. Because the tilt is fixed by velocity data, the CMB quadrupole is not asked to carry it; the observational degeneracy of (4) is broken by using two different data sets for the two sectors.
5. **A local/global claim firewall.** Local (depth-resolved) and global (bulk) statements are kept structurally distinct: the MIO depth-gap diagnostic G_F explicitly forbids the string “global tilt” in its metadata (`http/mio/formalism/isotropy_gap.py:_FORBIDDEN_G_METADATA_TERMS`), and G_F is documented as a depth-local bookkeeping quantity, not evidence for a global tilt. A local boost and a global tilt cannot be conflated through the diagnostic layer.

The E2E null (the matched Planck simulation ensemble; see the K1 download guide) is the calibration that turns safeguard 3 quantitative: any residual frame / boost quadrupole present in the observed map but absent from the simulations raises the look-elsewhere score, which is exactly where such a systematic should appear.

5 The comparator covariance is not diagonal

Let the data be d with covariance C and model $d = Dg$ (locally linear response). The Fisher information on g is

$$F = D^T C^{-1} D, \quad \text{Cov}(\hat{g}) \gtrsim F^{-1}. \quad (5)$$

With the exact D carrying the β^2 entry (4), the block between the two velocity-like sectors is

$$F_{\{\Sigma^2, \Omega_{\text{tilt}}\}} = \begin{pmatrix} F_{\Sigma^2 \Sigma^2} & F_{\Sigma^2 \Omega_{\text{tilt}}} \\ F_{\Sigma^2 \Omega_{\text{tilt}}} & F_{\Omega_{\text{tilt}} \Omega_{\text{tilt}}} \end{pmatrix}, \quad F_{\Sigma^2 \Omega_{\text{tilt}}} \propto \beta^2 \neq 0. \quad (6)$$

The off-diagonal is the analytic statement of the degeneracy: at fixed data, an increase in the inferred Ω_{tilt} can be partly absorbed by a decrease in Σ^2 along the eigenvector of (6) with the smaller eigenvalue. The leading-order code map Table 1 sets $F_{\Sigma^2 \Omega_{\text{tilt}}} = 0$ *by construction*; that is precisely the approximation the toy test stresses, and the reason the pipeline relies on the structural safeguards of §4 rather than on the diagonal assumption.

6 The two formalism upgrades, made precise

Both remedies proposed in the toy test are correct and slot directly onto the existing response-design / identifiability machinery.

6.1 Upgrade 1 — a kinematic projection operator

Define the deprojected (orthogonalised) shear estimator

$$\widetilde{\Sigma}^2 = \Sigma^2 - \alpha (\Omega_{\text{tilt}})^2, \quad (7)$$

which subtracts the boost-sourced quadrupole *before* Σ^2 is read off. The coefficient α is fixed analytically by (2): since the kinematic quadrupole scales as β^2 and $\Omega_{\text{tilt}} \propto \beta$ (a tilt is linear in the velocity), $(\Omega_{\text{tilt}})^2 \propto \beta^2$ matches the leakage exactly, and α is the ratio of the boost-quadrupole response to the shear quadrupole response,

$$\alpha = \left. \frac{\partial C_2^{\text{kin}} / \partial (\Omega_{\text{tilt}})^2}{\partial C_2^{\text{shear}} / \partial \Sigma^2} \right|_{\text{leading order}}. \quad (8)$$

Operationally this is a projection $\mathcal{P}$ that removes, from the quadrupole, the component along the boost-quadrupole template $Y_{2m}(\hat{\mathbf{v}})$ (a $\hat{\mathbf{v}}$ -aligned $\ell = 2$ pattern), leaving the residual to Σ^2 . In the code this is the natural extension of the response design D : add the β^2 row of (4) and project it out, so that $\widetilde{\Sigma}^2 \rightarrow 0$ in a pure-bulk-flow sky.

6.2 Upgrade 2 — a coupled Fisher prior

Rather than subtracting, carry the coupling (6) into the sampler as a prior. Compute $F_{\Sigma^2 \Omega_{\text{tilt}}} \propto \beta^2$ analytically and hand the full 2×2 block to the MCMC/MCTS as the parameter-space metric. The chain then explores along the degeneracy eigenvector rather than treating Σ^2 and Ω_{tilt} as independent axes, and a bulk-flow-only sky produces a posterior that is elongated along “trade Σ^2 against Ω_{tilt} ” with Σ^2 consistent with zero, instead of a spurious $\Sigma^2 > 0$ point estimate. This is the covariance-aware version of the identifiability rank already computed in `egs3_graded_comparator.py:identifiable_rank`: the off-diagonal reduces the *effective* rank in the $\{\Sigma^2, \Omega_{\text{tilt}}\}$ block whenever β is large enough that the two responses overlap.

Relation between the two. Upgrade 1 is a point-estimate deprojection; upgrade 2 is its Bayesian dual. They agree at leading order: the projection (7) is the maximum-likelihood solution of the coupled Fisher block (6) after marginalising Ω_{tilt} .

7 The “Leaky Universe” test, answered

Setup: a spatially flat sky ($\Omega_k = 0$) with no shear ($\Sigma^2 = 0$) and a single injected bulk flow Ω_{tilt} of 300 km s^{-1} ; Planck-level temperature noise; CF4-level (30%) distance errors; severe cosmic variance at $\ell \leq 30$.

Q1. Does a naive pipeline report a false $\Sigma^2 > 0$? Yes — a naive estimator that equates all quadrupole power with shear will pick up the β^2 kinematic quadrupole of (2) and report $\Sigma^2 > 0$. This is a real effect, correctly identified. It is the reason the diagonal map Table 1 is labelled leading-order-only.

Q2. Does the *current* pipeline report a false detection? No. It does not report a shear amplitude at all: Σ^2 is *partial* (look-elsewhere only), so there is no amplitude to be falsely inflated (safeguard 2). The $\ell = 1$ boost is excluded (safeguard 1). The boost-aligned quadrupole instead registers on the `axis_to_cmb_dipole_deg` statistic (safeguard 3), which is calibrated against the null — i.e. the pipeline would flag “low- ℓ axis aligned with the dipole apex”, the honest description of a boost, not “shear detected”. And the tilt itself is measured from CF4 velocities (safeguard 4), so the quadrupole is never asked to supply it.

- Q3. Is $\text{Cov}(g)$ diagonal? No** — (6): $F_{\Sigma^2\Omega_{\text{tilt}}} \propto \beta^2 \neq 0$. The code’s leading-order map sets it to zero *by construction*, which is why the pipeline leans on the structural safeguards rather than the diagonal assumption. Making the off-diagonal explicit is upgrade 2.
- Q4. Are the two proposed fixes valid? Yes, both.** The kinematic projection operator (7) with the analytic α deprojects the leakage so that $\widetilde{\Sigma}^2 \rightarrow 0$ in a pure-bulk-flow sky; the coupled Fisher prior (6) makes the sampler aware of the Σ^2 – Ω_{tilt} degeneracy. They are point-estimate and Bayesian versions of the same deprojection and attach to the existing response-design and identifiability code.

8 Claim envelope

This note is diagnostic-only. Ω_{tilt} is a model-independent kinematic descriptor (a bulk flow), not evidence of anisotropic geometry; Σ^2 is not reported as a measured amplitude; the local boost and the global tilt are kept distinct by construction. Nothing here identifies a Bianchi family, detects a geometry, validates a native low- ℓ solver, or promotes any scalar into a model-selection statement. The kinematic-quadrupole deprojection of §6 is the recommended formalism upgrade; until it is in the low- ℓ statistics, the separation is carried by the five structural safeguards of §4, and Σ^2 stays **partial**.